NEXUS TELECOM
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Executive Abstract: The Mandate for Action A data-driven diagnosis of critical churn vulnerabilities and a proposal for actionable strategies to secure customer value. This analysis pinpoints vulnerabilities in contract type and payment methods as primary drivers of attrition and provides a phased strategic plan to mitigate these risks and secure long-term revenue.

CUSTOMER CHURN ANALYSIS

→Strategic Recommendations for Retention

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Customer Churn Analysis: Strategic Retention Imperatives

❖ Problem Statement: Quantifying and Halting Critical Churn

The company faces a significant and quantified customer attrition risk (an overall churn rate). This analysis is mandated to statistically isolate the highest-leverage risk factors, define actionable retention targets, and establish a profit-driven implementation strategy to protect Customer Lifetime Value (CLV).

1. Executive Summary: The Retention Opportunity

This analytical report presents a comprehensive exploration of customer churn patterns within the Telco dataset. The objective is to statistically isolate the highest-impact factors driving customer attrition to guide immediate and long-term retention initiatives.

The analysis confirms a critical churn rate of **26.54%** across the customer base. The most severe attrition risks are concentrated among customers with **Month-to-month contracts (42.71% churn rate)**, those utilizing **Electronic check payments (churn rate)**, and customers in their **early tenure (up to 50% churn in the first year)**.

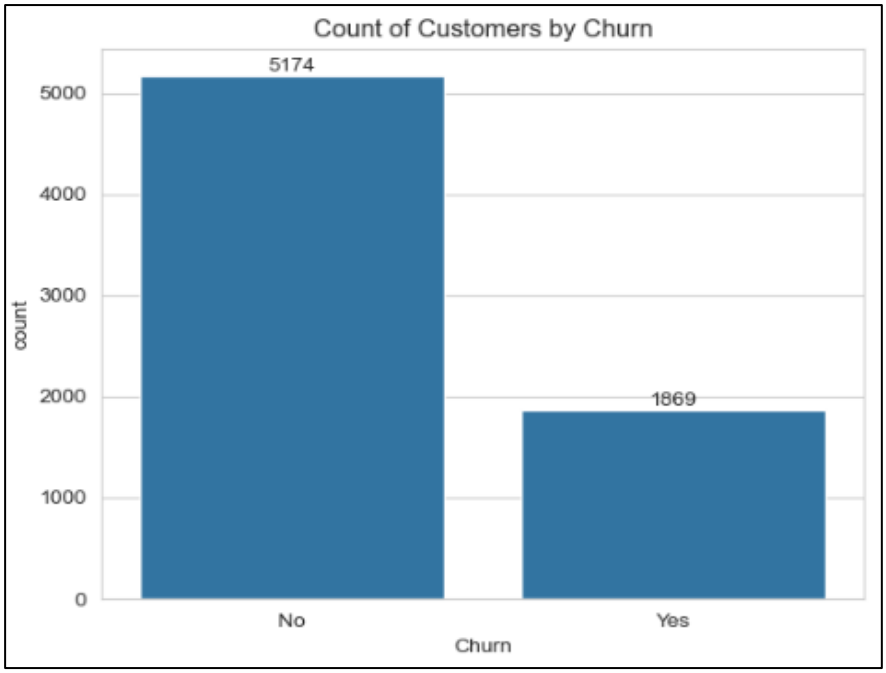
Key Statistical Findings:

Feature	Highest Risk Category	Churn Rate	Total Count	Churn Count
Contract	Month-to-month	**42.71%**	3,875	1,655
Payment Method	Electronic check	**45.29%**	2,365	1,071
Internet Service	Fiber optic	**41.89%**	3,096	1,297
Online Security	No	**41.77%**	3,498	1,461
Citizen	Senior	**41.00%**	1142	476

2. Methodology and Data Integrity

The analysis was executed on a complete dataset of 7,043 customer records. All data cleaning was performed successfully:

1. **Data Type Conversion:** The 'Total Charges' column was converted from 'object' to 'float' after correctly imputing **0** for **11** records where `**tenure = 0**`.
2. **Quality Check:** Confirmed **0** missing values and **0** duplicate records, ensuring a high-integrity basis for all statistical inference.
3. **Overall Churn:** The calculated overall churn ratio is **1869/7043 = 26.54%**



3. High-Impact Churn Drivers

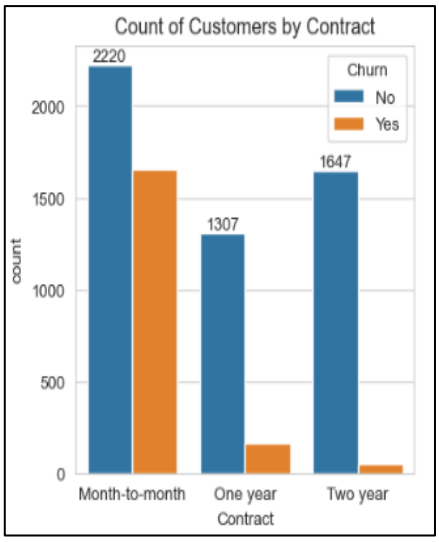
The following section details the most significant statistical associations between customer attributes and attrition risk.

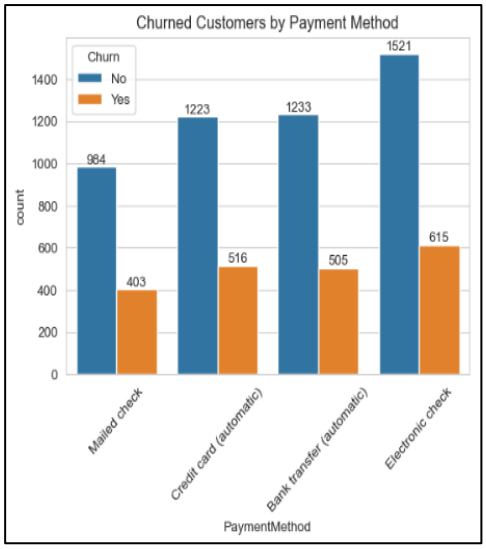
3.1 Contract and Payment Instability (Primary Risks)

Finding: Contract type and payment mechanism represent the two greatest, most statistically isolated risks for churn.

Contract Type	Churn Rate	Status	Payment Method	Churn Rate	Status
Month-to-month	**42.71%**	**Critical**	**Electronic check**	**45.29%.**	**Critical**
One year	**11.00%**	Low	**Credit card (automatic)**	**15.00%.**	Low
Two years	**3.00%**	Very Low	**Bank transfer (automatic)**	**15.00%.**	Low
			Mailed check	**18.00%.**	Moderate

- **Contract Implication:** Customers lacking a term commitment are 14 **times** more likely to churn than those on a two-year contract. The volume of churned month-to-month customers (1,655of 1,869) drives **88.5%** of all attrition.
- **Payment Implication:** The churn rate of **45.29%** among Electronic check users (totalling churned customers) is an extreme vulnerability, suggesting systemic friction, security concerns, or poor UX associated with this specific method.





3.2 Tenure and Engagement (Temporal Risk)

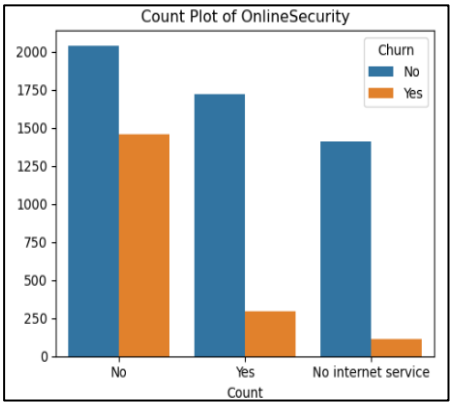
Finding: Churn is highest during the customer acquisition and onboarding phase, decreasing dramatically with loyalty.

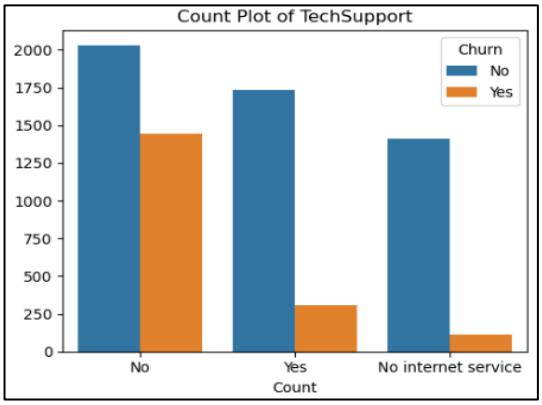
- **Early Attrition:** Customers with **less than one year of tenure** exhibit a churn rate approaching **50%.**
- **Stabilization:** Churn drops to **35%** for the 1-3 year bracket, stabilizing at just **15%** after the third year.
- **Implication:** Retention efforts must be front-loaded, recognizing the first year as the primary battleground for customer loyalty.

3.3 Service Failures and Demographics (Structural Risks)

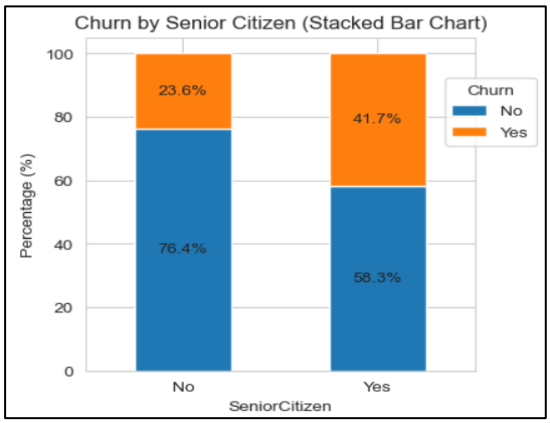
Finding: Core service dissatisfaction and unaddressed demographic needs contribute significantly to attrition.

- **Fiber Optic Service:** Customers using the Fiber Optic service display a high churn rate of **41.89%** (**1,297** churned customers). This is almost identical to the risk of having no foundational security or support services.
- **Missing Add-ons:** A customer lacking **Online Security (41.77% churn)** or **Tech Support (41.64% churn)** is at a greater churn risk than the overall group average. This indicates the high protective value of these services.





- **Senior Demographic:** Senior Citizens have a churn rate of approximately **41.7%**, far exceeding the **23.6%** rate of the non-senior cohort. This suggests a failure to provide tailored assistance or pricing sensitivity within this segment.



4. Strategic Recommendations: The Path to Profitability

The following three-phased strategic plan is designed to directly address the highest-risk churn drivers, leveraging the identified statistical thresholds to maximize retention and drive increased Customer Lifetime Value (CLV).

4.1 Phase I: Immediate Risk Mitigation & Stabilization (0-6 Months)

Strategic Priority	Key Initiative	Implementation Detail & Target Metric
Contract Conversion	**Term Commitment Incentive (TCI) Program:** Offer a **10%** monthly fee discount *plus* a free upgrade (e.g., free Online Security for 6 months) for Month-to-month customers who commit to a 1-year or 2-year contract.	**Target:** Convert **50%** of Month-to-month users, effectively reducing the **42.71%** churn pool by half.
Payment Method De-Risking	**Payment Migration Campaign:** Launch a focused campaign encouraging Electronic check users to switch to automated methods (Bank Transfer/Credit Card) via a 20 bill credit incentive.	**Target:** Migrate **30%** of Electronic check users, aiming to drop the **45.29%** churn rate to the industry average of **15%-18%**
Early Tenure Engagement	**Proactive Onboarding Check-in (POC):** Implement mandatory customer success outreach (email/SMS/call) at the 30-day and 60-day mark to resolve initial service friction.	**Target:** Reduce churn in the critical first year (**50%** risk segment) by 5-7 percentage points.

4.2 Phase II: Service Integrity & Value Enhancement (6-12 Months)

Strategic Priority	Key Initiative	Implementation Detail & Target Metric
Fiber Optic Service Audit	**Root Cause Analysis (RCA) & Infrastructure Fix:** Conduct a full diagnostic of the Fiber Optic network and technical support experience. Implement fixes to directly address the **41.89%** churn rate.	**Target:** Reduce Fiber Optic churn to the DSL level (**20%**) within 12 months.
Essential Protection Mandate	**"Security First" Bundling:** Integrate a non-removable, base-level Security/Tech Support package into all new Fiber Optic and DSL contracts.	**Target:** Increase add-on coverage for low-tenure customers by **40%**, leveraging the protective effect of these services against the **41%** risk of customers without them.

4.3 Phase III: Segmented Loyalty Programs (Long-Term Profitability)

Strategic Priority	Key Initiative	Implementation Detail & Target Metric
Senior Demographic Loyalty	**"Silver Tier" Support & Pricing:** Introduce a specialized Senior plan featuring simplified billing, priority access to human-led tech support, and a **15%** discount.	**Target:** Bring the Senior Citizen churn rate down from **41%** to be aligned with the non-senior baseline (**26%**).
CLV Maximization	**Multi-Year Renewal Rewards:** Create a dynamic retention model offering exclusive benefits (e.g., free modem upgrade, rate lock) to customers approaching the 1-year and 2-year contract expiration dates.	**Target:** Maintain loyalty stabilization (**15%** sub- churn) beyond the 3-year tenure mark, maximizing long-term revenue streams.

❖ Executive Closure

We wish to express our sincere gratitude for the opportunity to present this analysis**.** This report concludes the formal findings of the Customer Churn Study, and we confirm that all data presented is accurate and provides the necessary mandate for the strategic implementation phases outlined in **Section 4**.

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- **Prepared for:** The Executive Stakeholders
- **Date:** October 8, 2025
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